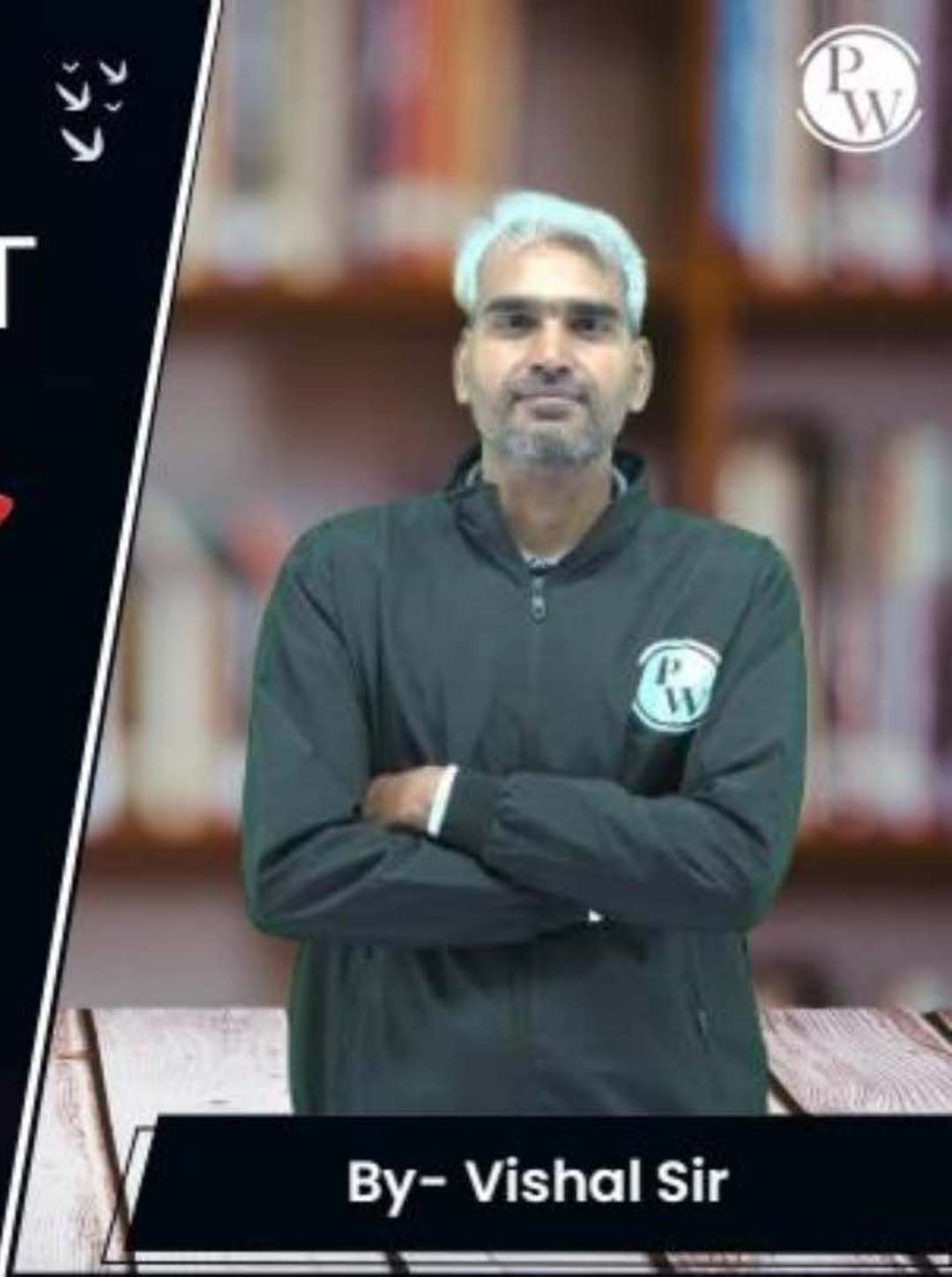
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 12



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Recap of Previous Lecture



Topic Properties of decomposition

Topic Dependency preserving decomposition

Slide

Topics to be Covered



- Topic: Lossless join decomposition
- Topic: Projection, Selection, and Cross product
- Topic: Natural join
- Topic: Lossless Natural Join

Lossless Join decomposition

Read about following relational algebra operations.

1. Projection (π)
2. Selection (σ)
3. Cross Product ($\times$)
4. Natural Join ($\bowtie$)

Properties of decomposition

1) Dependency Preserving decomposition

Let R be the relational schema with FD set F is decomposed into sub-relations $R_1, R_2, R_3 \ldots R_m$ with FD sets $F_1, F_2, F_3, \ldots F_m$ respectively

i.e.



then in general

$F_1 \cup F_2 \cup F_3 \cup \ldots \cup F_m \subseteq F$ { Always true }

if, $F_1 \cup F_2 \cup F_3 \cup \ldots \cup F_m = F$, then dependency Preserving

if, $F_1 \cup F_2 \cup F_3 \cup \ldots \cup F_m \subset F$, then not a dep. preserving decomposition

$F_1 \cup F_2 \cup F_3 \cup \ldots \cup F_m \supset F$ Not possible

2) Lossless Join decomposition

Let R be the relational schema with FD set F is decomposed into sub-relations $R_1, R_2, R_3 \ldots R_m$ with FD sets $F_1, F_2, F_3, \ldots F_m$ respectively

i.e.



then, in general,

$R_1 \bowtie R_2 \bowtie R_3 \bowtie \ldots \bowtie R_m \supseteq R$

if $=$, then lossless Join decomposition

if $\supset$, then lossy Join decomposition

if $\subset$, Not possible



Topic : Lossless Join decomposition

If we decompose a relation R with FD set F into sub-relations R1, R2,.....,Rn with FD sets F1, F2,.....,Fn respectively, then for this decomposition to be called lossless join decomposition following property must hold true.

$$R1 \bowtie R2 \bowtie \ldots. \bowtie Rn = R$$



Topic : Lossless Join decomposition

Let relation R is decomposed into sub-relations R1, R2,.....,Rn

In general, $R1 \bowtie R2 \bowtie .... \bowtie Rn \supseteq R$

if, $R1 \bowtie R2 \bowtie .... \bowtie Rn = R$ then, Lossless join decomposition

if, $R1 \bowtie R2 \bowtie .... \bowtie Rn \supset R$ then, Lossy join decomposition

$R1 \bowtie R2 \bowtie .... \bowtie Rn \subset R$ (not possible)



Topic : Natural Join (⋈)

Natural Join(⋈) is a derived Relational Algebra operation, which is derived using three basic Relational Algebra operation

- Projection (π)
- Selection(σ)
- Cross Product ($\times$)



Topic : Projection (π)

- It is used to project the column data from a relation (table) based on the attributes specified with projection operation.

Eg: Consider the relational schema $R(A_1, A_2, A_3, A_4, A_5, A_6)$

$$\pi_{A_1, A_3, A_4}(R)$$

Resulting schema will contain only three attributes, i.e., A_1, A_3 & A_4

List of attributes required in O/P

Name of relation

Syntax: $\pi_{\text{List of attributes}}(\text{Relation_name})$

eg:

Enroll

Sid	Cid	Branch
S_1	C_1	CS
S_1	C_2	CS
S_2	C_2	IT
S_3	C_3	CS

$\pi_{Sid, Cid, Branch}(Enroll)$ $\equiv$ (Enroll)

o/p of this R.A query will be the Complete "Enroll" table

If we do not use the Projection operation, then query will output all the attributes (Columns) from the relation

Retriere Sids of all the Students who enrolled for Some Courses:-

$\pi_{Sid}(Enroll) \Rightarrow$ o/p =

Sid
S_1
~~S_1~~
S_2
S_3

Sid
S_1
S_2
S_3

Relational algebra query will always Produce distinct tuples

eg:

Enroll

Sid	Cid	Branch
S_1	C_1	CS
S_1	C_2	CS
S_2	C_2	IT
S_3	C_3	CS

$\pi_{Cid, Branch}(Enroll) \Rightarrow$ o/p =

Cid	Branch
C_1	CS
C_2	CS
C_2	IT
C_3	CS

tuple wise they are distinct

Retriere Sids of all the Students who enrolled for Some Courses:-

$\pi_{Sid}(Enroll) \Rightarrow$ o/p =

Sid
S_1
S_1
S_2
S_3

Sid
S_1
S_2
S_3

Relational algebra query will always Produce distinct tuples



Topic : Selection(σ)

- It is used to select the tuples(records) from underlying relation based on the predicate condition specified with selection operation.

Syntax: $\sigma_{(Condition)}(Relation_name)$

{ If we do not specify the selection operation, then all the tuples from the relation will be selected }

eg:

Enroll

Sid	Cid	Branch
S_1	C_1	CS
S_1	C_2	CS
S_2	C_2	IT
S_3	C_3	CS

$\sigma_{(Cid = C_2)}(Enroll)$ ⇒ o/p =

Sid	Cid	Branch
S_1	C_2	CS
S_2	C_2	IT

No projection opn is used, so o/p will contain all three attributes

---

$\Pi_{Sid}\left(\sigma_{(Cid = C_1 \vee Branch = 'CS')}(Enroll)\right)$ ⇒ o/p =

Sid
S_1
S_3

o/p =

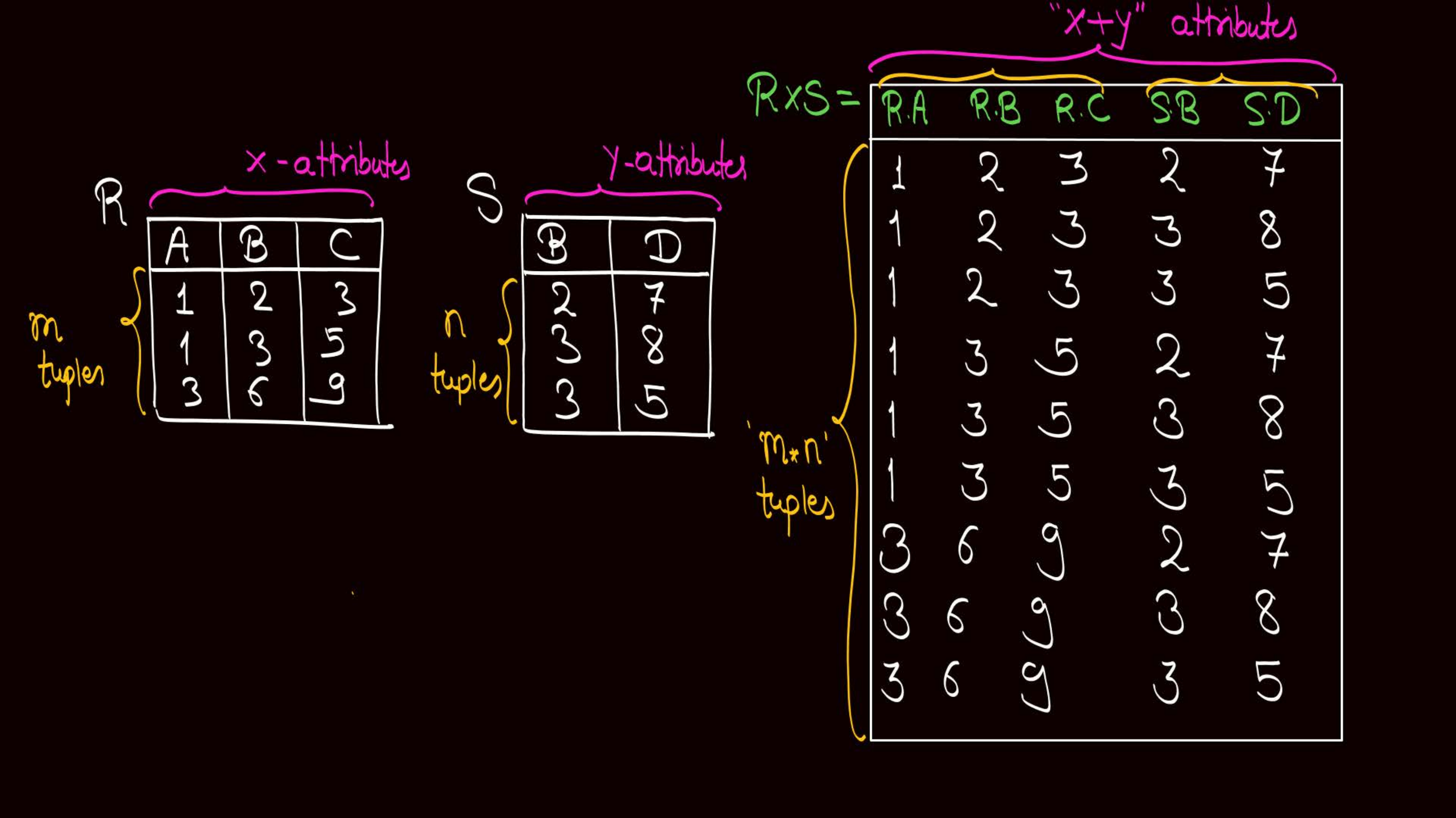
Sid	Cid	Branch
S_1	C_1	CS
S_1	C_2	CS
S_3	C_3	CS



Topic : Cross Product (×)

/ Catesian product / Cross Join / Cartesian Join

- Cross-product is a binary operation. Let R and S are any two relation, then cross product R × S will result in all attributes of R followed by all attribute of S with all possible combinations of tuples from R and S. {i.e; each tuple of R will be combined with each tuple of S}


x-attributes
R
A
B
C
1
2
3
1
3
5
3
6
9
m tuples
y-attributes
S
B
D
2
7
3
8
3
5
n tuples
"x+y" attributes
R x S =
R.A
R.B
R.C
S.B
S.D
1
2
3
2
7
1
2
3
3
8
1
2
3
3
5
1
3
5
2
7
1
3
5
3
8
1
3
5
3
5
3
6
9
2
7
3
6
9
3
8
3
6
9
3
5
'm*n' tuples



Topic : Natural Join (⋈)

- Natural join(⋈) is a derived relational algebra operation which is derived using cross product, selection and projection as follows:

Let R and S are any two relations then,

R⋈S = Step-1: Obtain "R × S"

Step-2: Select the tuples from "R × S" based on the equality condition on all common attributes of R and S.

Step-3: Project distinct attributes from the result of step-2.

R

A	B	C
1	2	3
1	3	5
3	6	9

S

B	D
2	7
3	8
3	5

$R \bowtie S = \Pi_{A,B,C,D}\left(\sigma_{R.B=S.B}(R \times S)\right)$

Common attribute is Projected only once

$R \times S =$

R.A	R.B	R.C	S.B	S.D
1	2	3	2	7
1	2	3	3	8
1	2	3	3	5
1	3	5	2	7
1	3	5	3	8
1	3	5	3	5
3	6	9	2	7
3	6	9	3	8
3	6	9	3	5

$R \bowtie S =$

A	B	C	D
1	2	3	7
1	3	5	8
1	3	5	5

Note:

Let, $R(A, B, C)$ & $S(B, D)$

$$R \bowtie S = \pi_{A,B,C,D}\left(\sigma_{R.B=S.B}(R \times S)\right)$$

Let $R(A, B, C)$ & $S(B, C, D, E)$

$$R \bowtie S = \pi_{A,B,C,D,E}\left(\sigma_{(R.B=S.B \wedge R.C=S.C)}(R \times S)\right)$$

Note: $R(A, B, C)$ & $S(D, E)$ { No Common attribute b/w R & S }

$$R \bowtie S = \pi_{A,B,C,D,E}\left(\underbrace{\quad}_{\text{No Common attribute between R\&S}}(R \times S)\right) \equiv R \times S$$

∴ No Selection Condition

⇓

Hence all tuples of R×S will be selected

finally project all the attributes of R×S

∴ Result will be same as "R×S"

Note:- If there is no common attribute b/w R & S, then output of "R ⋈ S" will be same as "R x S".

i.e., If there is no common attribute between R & S, then R ⋈ S will degenerate into R x S

eg: let

R =

A	B
1	3
2	5
6	9

S =

B	C
2	8
4	6
6	9

What will be o/p of R ⋈ S =

A	B	C

there will be no tuple in o/p

∴ o/p of R ⋈ S will be Empty relation



Lossless Natural Join.

* Consider the following relational table.

R

A	B	C
1	1	1
2	1	2
3	2	1

Case (1) If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(BC)$, then check whether the decomposition is lossless Join decomposition or not?

R_1=

A	B
1	1
2	1
3	2

R_2=

B	C
1	1
1	2
2	1

$R_1 \bowtie R_2$=

A	B	C
1	1	1
1	1	2
2	1	1
2	1	2
3	2	1

tuple from R

Extra tuples { Known as Spurious tuples }

$\therefore R_1 \bowtie R_2 \supset R$

Hence, Lossy decomposition

H.W.

Consider the following relational table.

R

A	B	C
1	1	1
2	1	2
3	2	1

② If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(AC)$, then check whether the decomposition is lossless Join decomposition or not?

R_1=

A	B
1	1
2	1
3	2

R_2=

A	C
1	1
2	2
3	1

$R_1 \bowtie R_2$=

A	B	C

H.W.

Consider the following relational table.

R

A	B	C
1	1	2
2	1	2
3	2	1

③ If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(BC)$, then check whether the decomposition is lossless Join decomposition or not?

R_1=

A	B
1	1
2	1
3	2

R_2=

B	C
1	2
2	1

$R_1 \bowtie R_2$=

A	B	C



2 mins Summary

- **Topic** Lossless join decomposition
- **Topic** Projection, Selection, and Cross product
- **Topic** Natural join
- **Topic** Lossless Natural Join

THANK - YOU